

$N=5$

Row=1 space = $N-1$ star = 1

while (row ≤ n) {

1, space

1, star

// next line prep
star++
star += 2
space--

}

Ques 10:
n = 5

```
* * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
```

$N=5$ Row=1 Star=2xN-1, space=0

while (row ≤ n) {

1, space

1, star

// next line prep

Row++

space++

star += 2

space--

Ques 12:
n = 5

```
*
 * *
 * * *
 * * * *
 * * * * *
```

Ques 12:
n = 5

Row=x Star=x, space=x

3 2 5 3 2 5

```
*
 * ! *
 * ! * ! *
 * ! * ! * ! *
```

```
--- *
--- * ! *
--- * ! * ! *
```

j=5 j=2,7

①

Row=1 space = 2xN-3, star=1

while (row ≤ n) {

1, star i ≤ star

1, space i ≤ star

1, star k ≤ star

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = n - 1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print(" ");
            i++;
        }
        // star
        int j = 1;
        while (j <= star) {
            System.out.print("* ");
            j++;
        }
        // next line ki prep
        row++;
        System.out.println();
        star += 2;
        space--;
    }
}
```

Ques 21:
n = 5

```
*
 * *
 * * *
 * * * *
 * * * * *
```

2xN-1-2

2xN-3

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = 2 * n - 3;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // space
        int j = 1;
        while (j <= space) {
            System.out.print(" ");
            j++;
        }
        // star
        int k = 1;
        while (k <= star) {
            System.out.print("* ");
            k++;
        }
        // next row ki prep
        row++;
        System.out.println();
        space -= 2;
        star++;
    }
}
```

Row=x Star=x Space=x

2 3 4 5 2 3 4 5 2 3 4 5

```
* - - - - *
* * - - - *
* * * - - *
* * * * - *
* * * * * *
```

i if (row == n) 2

k=2

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = 2 * n - 3;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // space
        int j = 1;
        while (j <= space) {
            System.out.print(" ");
            j++;
        }
        // star
        int k = 1;
        while (k <= star) {
            System.out.print("* ");
            k++;
        }
        // next row ki prep
        row++;
        System.out.println();
        space -= 2;
        star++;
    }
}
```

$N=5$ Row=x Star=1, space=7

2 3 4 5 2 3 4 5 2 3 4 5

```
* - - - - *
* * - - - *
* * * - - *
* * * * - *
* * * * * *
```

2 3 4 5

Ques 22:
n = 5

```
* * * * *
 * * * *
  * * *
   * *
    *
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = n;
    int space = -1;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // space
        int j = 1;
        while (j <= space) {
            System.out.print(" ");
            j++;
        }
        // star
        int k = 1;
        while (k <= star) {
            System.out.print("* ");
            k++;
        }
        // next row ki prep
        row++;
        space += 2;
        star--;
        System.out.println();
    }
}
```

$N=5$ Row=1, Star=N, space=-1

2 3 4 5 2 3 4 5 2 3 4 5

```
* * * * *
* * * *
* * *
* *
*
```

Ques 7:
n = 5

```
* * * * *
 * * * *
  * * *
   * *
    *
```

Row=1 Star=N

while (row ≤ n) {

int i=1

while (i ≤ star+1)

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = n;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // Next row ki prep
        row++;
        System.out.println();
    }
}
```

$N=5$ Row=x, Star=N

2 3 4 5 2 3 4 5 2 3 4 5

```
* * * * *
* * * *
* * *
* *
*
```

Row=x Star=N

i=1, i=star

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = n;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // Next row ki prep
        row++;
        System.out.println();
    }
}
```

$N=5$ Row=x, Star=N

2 3 4 5 2 3 4 5 2 3 4 5

```
* * * * *
* * * *
* * *
* *
*
```